• **DRIVEABILITY—ELECTRIC VEHICLE (EV)—DIAGNOSTIC TROUBLE CODE (DTC) C1838—BATTERY CHARGING DIAGNOSTICS SERVICE TIPS** • **ELECTRICAL—ELECTRIC VEHICLE (EV)—DIAGNOSTIC TROUBLE CODE (DTC) C1838—BATTERY CHARGING DIAGNOSTICS SERVICE TIPS**	**Article No. 01-10-9**

FORD: 1998-2001 RANGER

ISSUE

Some Electric Vehicle (EV) Rangers may exhibit a "No-Charge" condition. There are several possible causes why the Battery Control Module (BCM) might not allow charging (as indicated by the OP_STATES). A "Charging System Fault" (C1838) does not necessarily indicate a charger failure. The procedure below will help to pinpoint a No-Charge condition using previously stored data in the BCM.

ACTION

If the technician receives Diagnostic Trouble Code (DTC) "Charging System Fault" (Code C1838), further diagnosis will need to be performed. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

Attempt to verify the No-Charge condition. There may be several causes for this condition. Of those causes related to the charger, there are several that point beyond the charger to the wall box or to other system conditions and do not directly indicate a problem with the charger. Only a few indicate real failures in the charger itself.

If the technician has the opportunity to retrieve data while the vehicle is still On-Plug, there are a few other items that may assist the technician in repairing the No-Charge condition. These would be:

- "EV Operating States" (BCM_OP)
- "Charging Capacity" (CHG_CAP)
- "Average Temperature" (AVGTEMP)
- "Maximum Temperature" (MAXTEMP)
- "Charging Level Output Status" (CHG_LVL)
- "Auxiliary Current" (AUX_CNT)
- "Parallel Combination of Bus Leakage Resistance" (LEAKRES)

Other items that would be of interest would be environmental and usage information. These items cannot be retrieved from the BCM or the vehicle. Included would be:

- Temperature conditions of the vehicle
- Temperature conditions of the wall box
- Power quality
- Signal quality of the line capacity (measured by scope or chart recorder)
- Whether problems occur only with specific wall boxes, particular drivers or certain charge locations

Whatever additional information the technician can provide on the Repair Order in this area would be helpful.

At the time the technician receives the vehicle, the DTCs and the "Previous Operating States" (PREV_OP) should be read. The technician must do the following BEFORE placing the vehicle On-Plug, so the technician can read the condition that kept the vehicle from charging at the customer site.

1. Connect the New Generation Star (NGS) Tester tool when the vehicle arrives (BEFORE trying to put the vehicle on charge at the dealership).
2. Retrieve "Continuous Trouble Codes" (DTCs) from the BCM and record on Service Repair Order.
3. Select:
   - BCM (BAT CTRL MOD)
   - Select PID/DATA and select "Previous Operating States" (PREV_OP)

and record value on Service Repair Order.

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NOTE
THIS STEP MUST BE DONE PRIOR TO PUTTING THE VEHICLE ON-PLUG AT THE DEALERSHIP. IF THE PCS IS ATTACHED AFTER THE ORIGINAL FAULT HAS BEEN SET IN THE BCM (PREV_OP) PID, THEN THE OP-STATE THAT MAY HAVE HELPED TO DETERMINE CAUSE AT THE TIME OF THE ORIGINAL FAULT WILL BE LOST.

4. If DTC C1838 "Charging System Fault" is set (from Step 2), then read the following True/False items from the BCM PID/DATA and record any "True" items on Service Repair Order. Refer to the "Powertrain Diagnostics Routines Index" Section 100-09 of the Ranger EV Workshop Manual.
   - Charger Charging after being disabled (CHG_ABD)
   - Charger Over-Temperature Fault (CHG_OTF)
   - Charger Latch-up Fault (CHG_LUF)
   - Charger Under-Current Fault (CHG_UCF)
   - Charger Over-Current Fault (CHG_OCF)
   - Charger Over-Voltage Fault (CHG_OVF)
   - Charger Over-Amp Hours Fault (CHG_OAF)
   - Charger Line Capacity Fault (CGF_LCF)
   - Charger Mode Feedback Fault (CHG_MFF)
   - Charger Cord Circuit Fault (CHG_CCF)
   - Charger Cycling Fault (CHG_CYF)
5. After reading the above PIDS, the technician can place the vehicle On-Plug.
6. Using the NGS tool, monitor the "ECU Operating States" (BCM_OP) PID. This PID should be monitored. If the state is not one of the following:
   - "Charge Setup" (OP STATE 4)
   - "Charge (Current Mode)" (OP STATE 5)
   - or "Charge (Voltage Mode)" (OP STATE 6)

then record the "ECU Operating States" PID value on the Service Repair Order and retain for future reference.

If the vehicle charges properly for more than 30 minutes at the dealership, then the most probable cause of the No Charge condition was either a transient condition (overtemp) or a problem with the PCS unit at the customer's location.

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
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